

Glycolysis of Poly(ethylene terephthalate) Catalyzed by the Lewis Base Ionic Liquid [Bmim][OAc]

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ABSTRACT: The glycolysis of poly(ethylene terephthalate) (PET) was studied using 1-butyl-3-methylimidazolium acetate ([Bmim][OAc]) as a catalyst. The effects of temperature, time, ethylene glycol dosage, PET amount, and [Bmim][OAc] dosage on the glycolysis reaction were examined. The results revealed that [Bmim][OAc] has a PET conversion of 100% and a bis(2-hydroxyethyl)terephthalate (BHET) yield of 58.2% under the optimum conditions of 1.0 g of [Bmim][OAc] with 20 g of ethylene glycol in the presence of 3.0 g of PET at 190 °C after 3 h of glycolysis. The ionic liquid could be reused up to six times with no apparent decrease in the conversion of PET or yield of BHET. The pH plays a major role in explaining the proposed mechanism of glycolysis using the Lewis base ionic liquid [Bmim][OAc]. The kinetics of the reaction was first-order with an activation energy of 58.53 kJ/mol.

INTRODUCTION

Increased environmental awareness, legislative measures, and public demand for environmental sustainability are leading to an increased interest in plastics recycling. The amount of plastic produced has been increasing significantly each year, with uses including fiber, packing, containers, and building materials, among others.^{1,2} Poly(ethylene terephthalate) (PET) is a semicrystalline thermoplastic polyester with high chemical and impact resistance at room temperature. PET is extensively used in diverse applications, such as textiles, high-strength fibers, photographic films, and disposable soft-drink bottles, because of its excellent mechanical properties, chemical stability, safety, light weight, transparency, and air-tightness,^{3,4} resulting in a continuously growing stream of PET material.⁵ The total global PET consumption has risen to 54 million metric tons in 2010 and is expected to grow by 4.5% per year from 2010 to 2015.⁶ Thus, the effective recycling of PET waste is regarded as one of the most important ways of resolving “white pollution”, saving resources, and protecting the environment.^{7,8} Therefore, an ecologically safe method of recycling plastics, such as PET, remains an important scientific and societal goal.

At present, the conventional methods of PET recycling are mainly divided into physical methods and chemical methods. The physical methods of recycling produce some plastic products with inferior properties. However, the chemical methods can recycle PET waste into monomers that can be used as raw materials to produce virgin plastic products.⁹ Considering the quality of the recycled products, chemical methods are particularly attractive and have been studied widely. These methods mainly include methanolysis, hydrolysis, and glycolysis, which have been developed on commercial or pilot scales.^{10,11} Glycolysis is the most promising approach because of its advantages of low-volatility solvents and continuous production feasibility. The main product of the

glycolysis of PET is the virgin monomer bis(2-hydroxyethyl)-terephthalate (BHET), which can be used to make PET production units, textile softeners and unsaturated polyester resins.¹²

The glycolysis reaction is very slow in the absence of catalyst and is an extremely sluggish process.^{13,14} In recent years, many catalysts have been developed for the glycolysis of PET, such as metal acetates (Zn, Co, Pb, and Mn);^{8,15,16} titanium phosphate; solid superacids; and metal oxides, carbonates, and sulfates.^{17,18} However, these catalysts have several drawbacks, such as the need for high temperatures and high pressures, difficult separation of the catalyst from the depolymerized products, the occurrence of side reactions, and the impurity of the products.^{19,20} Therefore, it is necessary to develop new catalysts for the glycolysis of PET.

Recently, ionic liquids (ILs) have been used as environmentally friendly catalysts in the degradation of PET under mild conditions because of their adjustable physical and chemical properties,^{21–24} attracting the attention of scholars from various fields, such as synthesis, catalysis, separations, and electrochemistry.²⁵ The special properties of ionic liquids make them easier to separate from the solid glycolysis products. Nunes et al.²⁶ studied the depolymerization of PET in supercritical ethanol in the presence of the ionic liquid 1-butyl-3-methylimidazolium tetrafluoroborate ([Bmim][BF₄]). The use of the ionic liquid [Bmim][BF₄], in addition to supercritical ethanol, represented an extremely promising combination for PET depolymerization in a sustained way. Wang et al.¹² used an ionic liquid as a catalyst for the glycolysis

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of PET; however, the conversion of PET and selectivity toward BHET were very low, and the process was very slow. They also studied the glycolysis of PET using an Fe-containing magnetic ionic liquid, which exhibited higher catalytic activity than conventional ILs. However, the monomer produced was very easily stained by the Fe-containing ILs.²⁷

Liu et al.²⁸ reported that 1-butyl-3-methylimidazolium acetate ([Bmim][OAc]) exhibited excellent catalytic activity and reusability when used as a catalyst in the methanolysis of polycarbonate, compared with 1-butyl-3-methylimidazolium chloride ([Bmim][Cl]). In addition, Song et al.²⁹ studied the hydrolysis of polycarbonate using [Bmim][OAc] as a catalyst. They also reported that [Bmim][OAc] could act as an efficient catalyst in the methanolysis and hydrolysis of poly(lactic acid).^{30,31} However, we have found no reports on the use of [Bmim][OAc] in PET glycolysis. Using the acetate-based ionic liquid [Bmim][OAc] as a glycolysis catalyst could be a better option than using conventional catalysts with regard to high monomer yield, possibility of regeneration, ease of separation, and long shelf life. By recovering the catalyst, the danger of releasing harmful catalyst components to the environment is reduced. By reusing the catalyst, the process becomes more cost-effective and sustainable.

The aim of this investigation was to apply the acetate-based ionic liquid [Bmim][OAc] as a catalyst in the glycolysis of PET to avoid the negative environmental and health effects of chlorine-containing IL catalysts. To the best of our knowledge, the use of [Bmim][OAc] as a catalyst for the glycolysis of PET has yet to be published. In addition, a mechanism for the glycolysis of PET is proposed, and the role of pH in explaining the mechanism and kinetics of the reaction is discussed.

■ EXPERIMENTAL SECTION

Materials. PET pellets ($3.2 \times 2.8 \times 4$ mm) were purchased from Hangzhou Zhengnan Biological Technology Co., Ltd. The intrinsic viscosity of PET was measured in a 60:40 (w/w) phenol/1,1,2,2-tetrachloroethane solution at 25 °C and found to be 0.64 dL·g⁻¹. Using the Mark–Houwink method, the weight-average molecular weight (4.1×10^4 g·mol⁻¹) was calculated from this intrinsic viscosity. Analytical-grade ethylene glycol (EG), 1-butyl-3-methylimidazolium chloride ([Bmim][Cl]), 1-butyl-3-methylimidazolium bromide ([Bmim][Br]), and ethanol were obtained from Sigma-Aldrich. 1,1,2,2-Tetrachloroethane, potassium acetate, and phenol were obtained from Merck.

Synthesis of 1-Butyl-3-methylimidazolium Acetate ([Bmim][OAc]). 1-Butyl-3-methylimidazolium acetate ([Bmim][OAc]) was synthesized according to the procedures described in the literature.³²

General Procedure for the Glycolysis of PET. A 50 mL round-bottom three-neck flask equipped with a thermometer and a reflux condenser was loaded with 2.0 g of PET, 20.0 g of ethylene glycol, and a certain amount of catalyst. The glycolysis reaction was carried out under atmospheric pressure at reaction temperatures ranging from 150 to 190 °C for glycolysis times of 1–4 h. The flask was immersed in an oil bath at a specific temperature for the required time. When the glycolysis reaction was complete, the undepolymerized PET pellets were quickly separated from the liquid phase before the products precipitated. An excess amount of cold distilled water was used to wash the undepolymerized PET pellets, and the water was then mixed with the product fraction. The undepoly-

merized PET was collected, dried, and weighed. The conversion of PET is defined as

$$\text{conversion of PET (\%)} = \frac{W_0 - W_1}{W_0} \times 100\% \quad (1)$$

where W_0 represents the initial weight of PET and W_1 represents the weight of undepolymerized PET. Meanwhile, the glycolysis product mixture was vigorously agitated (the cold distilled water dissolved the remaining ethylene glycol, catalyst, and monomer) and then filtered. The collected filtrate was concentrated to approximately 150 mL on a vacuum rotary evaporator at 50 °C. The concentrated filtrate was stored in a refrigerator at 0 °C for 24 h. White crystalline flakes formed in the filtrate and were then separated and dried. This material was the bis(hydroxyethyl) terephthalate (BHET) monomer. The fraction insoluble in cold water was a mixture of the dimer and oligomers. The products were analyzed using X-ray diffraction (XRD), nuclear magnetic resonance (NMR) spectroscopy, electron ionization mass spectrometry (EI-MS), Fourier transform infrared (FTIR) spectroscopy, thermogravimetric analysis (TGA), and differential scanning calorimetry (DSC). The selectivity of the BHET monomer is defined as

$$\text{selectivity of BHET (\%)} = \frac{W_{\text{BHET}}/MW_{\text{BHET}}}{W_{\text{PET,D}}/MW_{\text{PET}}} \times 100\% \quad (2)$$

and the yield is defined as

$$\text{yield of BHET (\%)} = \frac{W_{\text{BHET}}/MW_{\text{BHET}}}{W_{\text{PET,I}}/MW_{\text{PET}}} \times 100\% \quad (3)$$

where $W_{\text{PET,D}}$, $W_{\text{PET,I}}$, and W_{BHET} refer to the weight of depolymerized PET, the initial weight of PET, and the weight of BHET, respectively. MW_{BHET} and MW_{PET} are the molecular weights of BHET (254 g·mol⁻¹) and the PET repeat unit (192 g·mol⁻¹), respectively.

Characterization. The IL and the main product were analyzed by ¹H NMR spectroscopy using an ECA 500 MHz instrument (JEOL, Tokyo, Japan) in deuterated dimethyl sulfoxide (DMSO-*d*₆) solution. Fourier transform infrared (FTIR) measurements were performed using a Nicolet IS-10 FTIR instrument with KBr discs. The mass spectra of the IL and the main product were obtained on an ISQ single-quadrupole MS (Thermo Scientific) instrument with electron ionization (EI). X-ray diffraction (XRD) patterns of the main product and PET material were recorded in the range of $2\theta = 4\text{--}80^\circ$ using a Philips powder diffractometer with Cu K α radiation ($\lambda = 0.154$ nm). The instrument was operated at 40 kV and 40 mA. The spectra were recorded at a scanning rate of $2^\circ/\text{min}$. Thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) of the main product and PET were performed using a SDTQ 600 system (TA Instruments, New Castle, DE) by heating from room temperature to 1000 °C at a rate of 10 °C/min in a nitrogen atmosphere. Scanning electron microscopy (SEM) images of PET before and after glycolysis were obtained using a JEOL 5410 system operating at 20 kV.

■ RESULTS AND DISCUSSION

Characterization of the Synthesized Ionic Liquid. The structure of the synthesized IL was confirmed by EI-MS and FTIR spectroscopy. The EI-MS spectrum of [Bmim][OAc] (Figure S1, Supporting Information) shows that the molecular

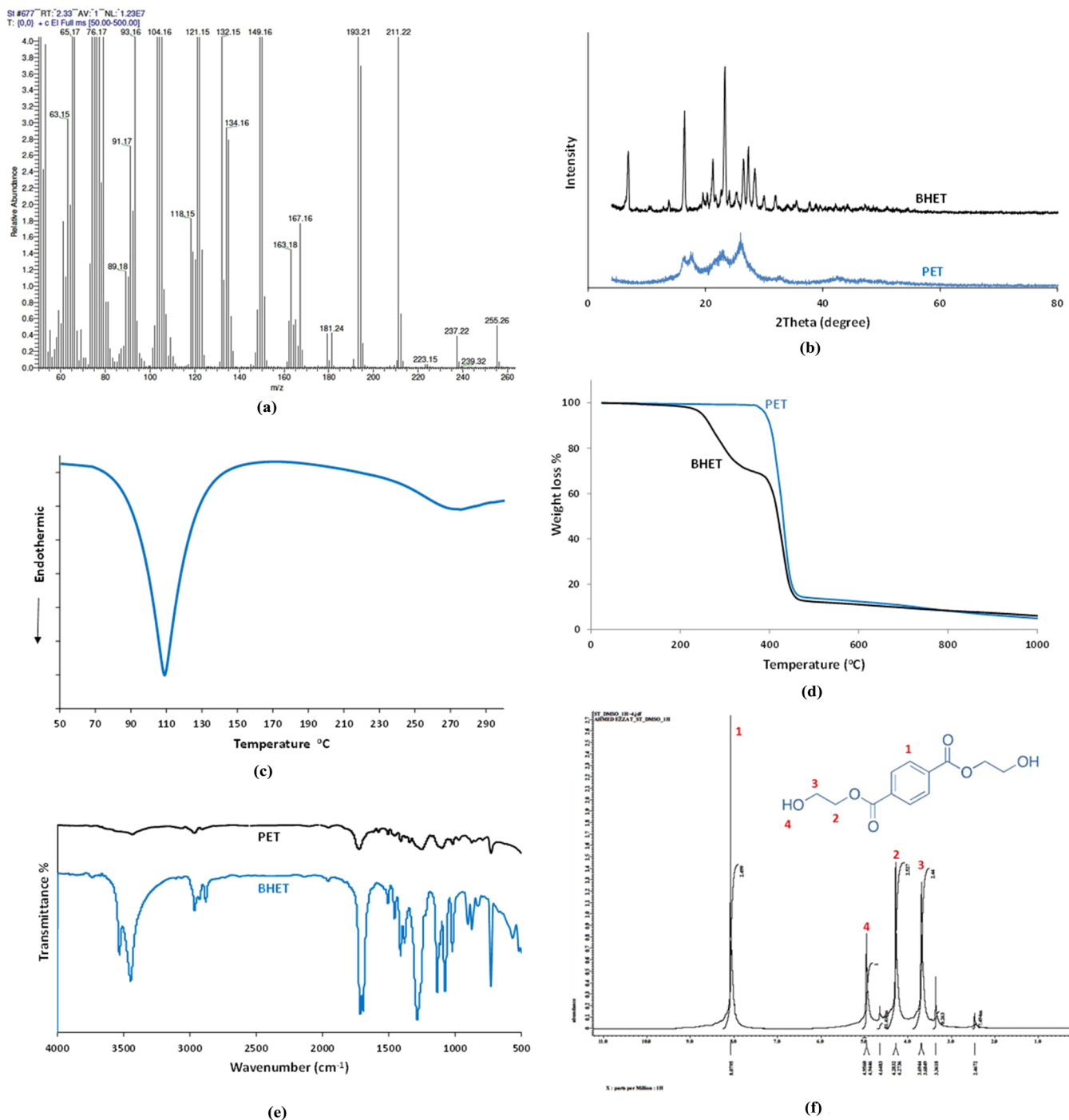


Figure 1. (a) EI-MS spectrum of BHET, (b) XRD spectra of BHET and PET, (c) DSC curve of BHET, (d) TGA curves of BHET and PET, (e) FTIR spectra of BHET and PET, (f) ^1H NMR spectrum of BHET.

weight of the synthesized ionic liquid was 198 g/mol, corresponding to the molecular weight of $[\text{Bmim}][\text{OAc}]$. Figure S2 (Supporting Information) compares the FTIR spectrum of the synthesized ILs with that of $[\text{Bmim}][\text{Cl}]$. The spectrum of $[\text{Bmim}][\text{OAc}]$ (Figure S2, Supporting Information) contains major peaks corresponding to acetate $\text{C}=\text{O}$ bond stretching ($1570\text{--}1579\text{ cm}^{-1}$), $\text{C}-\text{O}$ stretching ($1379\text{--}1403\text{ cm}^{-1}$), $\text{C}-\text{H}$ stretching ($2963\text{--}2966\text{ cm}^{-1}$), $\text{C}-\text{C}$ stretching (918 cm^{-1}), $\text{O}-\text{C}-\text{O}$ bending (650 cm^{-1}), imidazole $\text{C}-\text{N}$ stretching (1337 cm^{-1}), $\text{C}-\text{H}$ stretching (3104 cm^{-1}), side-chain $\text{C}-\text{H}$ stretching (2874 cm^{-1}), and

$\text{C}-\text{N}-\text{C}$ bending (623 cm^{-1}). The $\text{C}-\text{H}$ peaks between 3100 and 3200 cm^{-1} can be attributed to aromatic $\text{C}-\text{H}$ stretching, whereas those below 3000 cm^{-1} can be attributed to aliphatic $\text{C}-\text{H}$ stretching.^{33,34} The observed aromatic $\text{C}-\text{H}$ stretching is characteristic of $\text{C}-\text{H}\cdots\text{O}$ hydrogen bonds.³⁵

The ^1H NMR spectrum of $[\text{Bmim}][\text{OAc}]$ is reproduced in Figure S3 (Supporting Information) for illustration. The ^1H NMR spectrum (500 MHz , $\text{DMSO}-d_6$) included peaks at δ 9.86 (s, 1H, CH), 7.88 (s, 1H, CH), 7.80 (s, 1H, CH), 4.16 (t, 2H, NCH_2), 3.85 (s, 3H, NCH_3), 1.60 (s, 3H, CH_3CO), 1.71 (m, 2H, CH_2), 1.16 (m, 2H, CH_2), and 0.97 (t, 3H, CH_3).

Qualitative Analysis of the Main Degradation Product. To identify the structure of the main product in the degradation of PET in EG catalyzed by [Bmim][OAc], NMR, EI-MS, XRD, DSC, TGA, and FTIR characterizations were performed. The main product was also identified by measuring the melting temperature, which was found to be 110.9 °C. The EI-MS spectrum in Figure 1a shows that the molecular weight of the main product is 254 g/mol, which is the same as that of BHET. Figure 1b compares the XRD patterns of BHET and the PET material. PET exhibits a typical diffraction pattern because of the crystalline structure of this polyester, with broader diffraction peaks at $2\theta = 16.4^\circ$, 17.7° and 23.0° . The diffraction peaks of BHET are narrower and more intense than those of PET, which illustrates that the BHET obtained by this method has a high crystallinity and a different crystalline structure than PET. The DSC curve in Figure 1c shows a sharp endothermic peak at 110 °C. The melting onset temperature and peak temperature of BHET are 110.9 and 110 °C, respectively. The TGA curves of the PET material and BHET are illustrated in Figure 1d. The TGA curve of PET shows significant weight loss at 400 °C, which is attributed to the thermal decomposition of PET.^{36,25} The TGA curve of BHET exhibits two clear weight losses. The first is approximately 32% at an onset temperature of 240 °C due to the thermal decomposition of BHET. During the heating process in TGA, BHET repolymerizes to PET. The other loss is approximately 60% at an onset temperature 400 °C due to the thermal decomposition of PET produced by BHET thermal polymerization during TGA.³⁶ Figure 1e shows the FTIR spectra of the BHET product and PET material. The FTIR spectrum of the BHET monomer clearly exhibits OH bands at 3450 and 1135 cm^{-1} , a C=O stretching band at 1715 cm^{-1} , alkyl C–H bands at 2879 and 2954 cm^{-1} , and aromatic C–H bands at 1411–1504 cm^{-1} , which are all present in BHET.^{37,38}

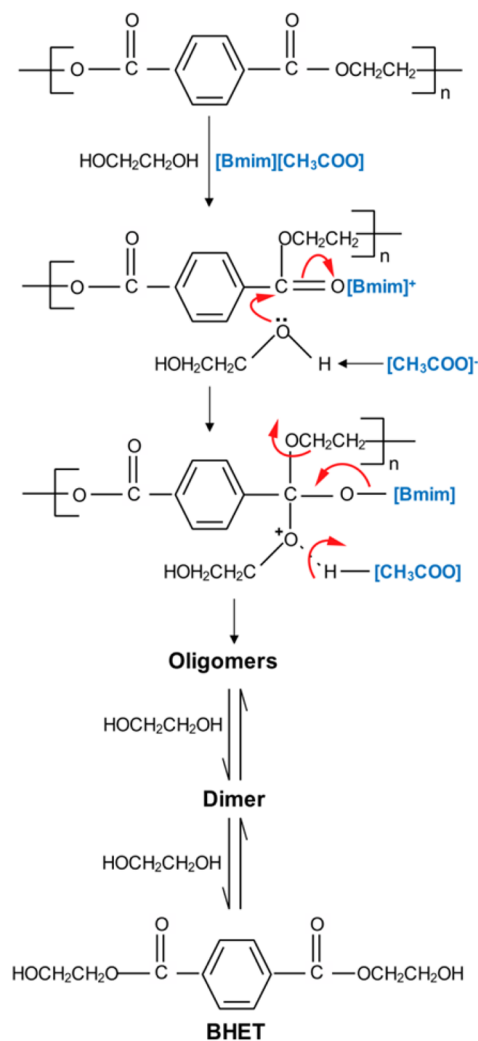
The ^1H NMR spectrum of BHET is presented in Figure 1f. The single signal at δ 8.1 ppm indicates the presence of the four aromatic protons of the benzene ring. The triplet at 4.3 ppm and the quartet at 3.7 ppm represent the methylene protons of $\text{COO}-\text{CH}_2$ and CH_2-OH , respectively. The triplet at δ 4.9 ppm is characteristic of the protons of the hydroxyl group. The ^1H NMR spectrum also agrees very well with that reported in the literature.³⁹

Effect of Catalysts on the Glycolysis of PET. The catalytic performances of various ionic liquid catalysts were examined in the glycolysis reaction of PET. As reported in Table S1 (Supporting Information), in the presence of the Lewis neutral ionic liquids [Bmim][Br] and [Bmim][Cl], no glycolysis of PET took place. However, when [Bmim][OAc] was used as a Lewis base catalyst under the same conditions, the conversion of PET was 75.1%, the selectivity of BHET was 61.9%, and the yield was 46.4%. This demonstrates that [Bmim][OAc] has excellent catalytic activity for the glycolysis of PET, as the degradation reaction occurred under relatively mild conditions with a higher reaction extent.

Effects of Reaction Conditions on the Glycolysis Reaction of PET. *Effect of Catalyst Concentration.* As shown in Table S1 (Supporting Information), the addition of the catalyst [Bmim][OAc] has a strong impact on conversion, selectivity, and yield in the glycolysis reaction of PET. Table S2 (Supporting Information) presents the influence of the catalyst amount on the glycolysis of PET. The conversion of PET increased sharply when the amount of [Bmim][OAc] was increased from 0.125 to 1.0 g. It reached 75.1% at 1.0 g and

then decreased to 41.5% at 3.0 g. This trend is the result of the increase in the catalyst content accelerating the reaction and shortening the equilibration time.⁴⁰ As more ionic liquid was added, more cations and anions became available to interact with the active centers of the reactants (shown in Scheme 1).

Scheme 1. Mechanism of the Glycolysis of PET Catalyzed by [Bmim][OAc]



Thus, more PET was glycolyzed. However, when the amount of [Bmim][OAc] exceeded 1.0 g, the conversion of PET decreased slightly, which might be due to the repolymerization of the glycolysis product.

As shown in Table S2 (Supporting Information), the selectivity and yield of BHET first increased with increasing catalyst weight, peaking at 61.9% and 46.4%, respectively, at 1.0 g of [Bmim][OAc]. They then varied insignificantly when the amount of catalyst exceeded 1.0 g. Presumably, when more [Bmim][OAc] is used, the BHET monomer is produced more quickly. BHET can then polymerize into dimers or oligomers as the glycolysis reaction proceeds, decreasing the amount of BHET monomer present.²⁷

Effect of Reaction Temperature. The effects of reaction temperature on the conversion of PET and the yield of BHET were investigated, and the results are reported in Table S3 (Supporting Information). The results reveal that the conversion of PET and the yield and selectivity of BHET

were greatly enhanced with increasing reaction temperature. Apparently, there is an interaction between EG and the catalyst that can activate the hydroxyl group in EG to enhance the degradation of PET.⁴¹ The high catalytic activity can mainly be attributed to the synergistic effect between the cation and anion of the IL catalyst. When the temperature was 150 °C, the PET conversion was only 11%, and there was no BHET yield. However, with increasing glycolysis temperature, the conversion of PET increased markedly, rapidly reaching 75% when the temperature was 180 °C. The conversion of PET reached a maximum value of 100% when the glycolysis temperature was 190 °C. Furthermore, the yield of BHET increased as the reaction temperature was increased. When the temperature was raised to 170 °C, the selectivity and yield of BHET were only 22% and 8.14%, respectively. The selectivity and yield of BHET reached maximum values of 61.9% and 46.4%, respectively, at 180 °C, beyond which the increasing tendency was retarded: Both declined to 41% when the temperature was increased to 190 °C. Therefore, the glycolysis temperature is very critical for the reaction.

Effect of Reaction Time. The effects of reaction time on the conversion of PET and the selectivity and yield of BHET were examined, and the results are presented in Table S4 (Supporting Information). It is obvious that the conversion of PET increased distinctly with increasing glycolysis time. When the reaction time was extended to 2.0 h, the PET conversion was 100%. Moreover, when the reaction time was more than 3.0 h, the yield of BHET decreased, and the conversion of PET remained at 100%. It was also found that the selectivity and yield of BHET reached maximum values when the reaction time was set at 3.0 h. The selectivity and yield of BHET decreased at prolonged reaction times. This behavior can be attributed to the polymerization of BHET carried out as a side reaction during the glycolysis of PET with increasing reaction time.¹² Thus, it can be concluded that, during the depolymerization process, PET was first depolymerized into oligomers. Next, the oligomers were converted into dimers and then into BHET monomers in the presence of ethylene glycol. With increasing glycolysis time, BHET would further polymerize into dimers and then into oligomers. This would account for the tendency of the selectivity of BHET to first increase and then decrease with increasing reaction time.¹²

Effect of Ethylene Glycol (EG) Dosage. The ethylene glycol (EG) dosage is also a very important parameter for the glycolysis reaction.¹⁶ The effect of ethylene glycol (EG) dosage on the glycolysis reaction is shown in Table S5 (Supporting Information). Under conditions of 1.0 g of [Bmim][OAc], a reaction temperature of 190 °C, 2 g of PET, and a 3-h reaction time, the conversion of PET and the selectivity and yield of BHET reached 100%, 53.5%, and 53.5%, respectively, at an EG dosage of 20 g. The PET conversion and BHET yield and selectivity all decreased as the EG dosage was increased to 25 g. This trend can be attributed to a decrease in catalytic consistency with increasing EG dosage.

Effect of PET Amount. The glycolysis reaction was investigated using 1.0–7.0 g of PET. Table S6 (Supporting Information) reveals that, under the reaction conditions of 1.0 g of [Bmim][OAc], 20 g of EG, a temperature of 190 °C, and a reaction time of 3 h, the conversion was 100%, even after the PET content had been increased to 5.0 g. Nevertheless, when the amount of PET reached 7.0 g, the conversion started to decline, as the PET could not be degraded completely. The selectivity and yield approached their maximum values when

the amount of PET was 3.0 g, after which they both started to decline: The selectivity dropped from 58.21% to 28.6%, and the yield decreased from 58.21% to 26%. It is obvious that, when the amount of PET exceeded 3.0 g, the yield of BHET decreased with increasing amount of PET. Thus, the glycolyzed portion was primarily dimer and oligomers, and the BHET content was very low.⁴⁰

Recycling of Residual EG and Catalyst and Reusability of [Bmim][OAc]. Considering the environmental friendliness and economics of the process, the recycling of the residual EG and the catalyst was carried out. One of the most attractive properties of ionic liquids is their good reusability when used as either a catalyst or a reaction medium.²⁶ The reusability of the ionic liquid [Bmim][OAc] in the glycolysis of PET was investigated, and Table S7 (Supporting Information) shows that the degradation efficiency did not decrease in the sixth reuse of the catalyst and residual EG. After BHET was filtered from the liquid phase, the residual EG and catalyst in the filtrate were recovered with a vacuum rotary evaporator at 65 °C, stored in a vacuum oven at 60 °C for 6 h, and then supplied with a certain amount of EG until the weight became equal to the initial weight of fresh EG and catalyst. The ionic liquid could be reused up to six times without any apparent decrease in the conversion of PET or in the selectivity and yield of BHET under the given conditions. It is well-known that the main factor affecting the reusable performance of ionic liquids is their stability under the reaction conditions. The loss of activity was perhaps due to the loss of [Bmim][OAc] during the workup. Therefore, [Bmim][OAc] is an effective and reusable catalyst for the glycolysis of PET.³²

Role of pH in Explaining the Glycolysis Mechanism Using [Bmim][OAc]. The role of pH could be deduced by monitoring the pH before and after the glycolysis reaction in the presence of the Lewis base ionic liquid [Bmim][OAc]. The pH values of the reaction medium before and after the reaction were found to be 8.0 and 6.5, respectively. The change in pH from 8.0 to 6.5 might be mainly due to the release of protons (H⁺) from ethylene glycol, which then interacted with the acetate anions to form acetic acid. Last, this weak monoprotic acid partially dissociated to give H⁺ and acetate anions. This explanation seems to be in good agreement with the proposed mechanism.

Degradation Mechanism for PET Glycolysis. There has been some debate about the mechanism of the degradation of PET. Researchers have reported that the degradation of PET results from the cleavage of the ester group in PET.^{27,42–44} In some studies, it was found that the degradation of PET occurred in the amorphous phase and at the chain folds on the crystal surface.^{45,46} This assumption could be further confirmed by scanning electron micrographs revealing the morphology of fresh PET and residual PET, as illustrated in Figure 2. The image presented in Figure 2a shows that the surface of fresh PET is relatively smooth. However, the surface of residual PET after 15, 30, and 60 min, as displayed in panels b–d, respectively, of Figure 2, seems to exhibit a porous structure, indicating the penetration of ethylene glycol into the PET pellets.

The SEM images show that the surfaces of undepolymerized PET pellets were covered by increasingly obvious cracks as the reaction time increased. Figure 2 indicates that the degradation of PET occurred place on the surfaces of the PET pellets, which conforms with the literature findings,^{41,42} and that the reaction rate increased with an increase in the exposed area resulting

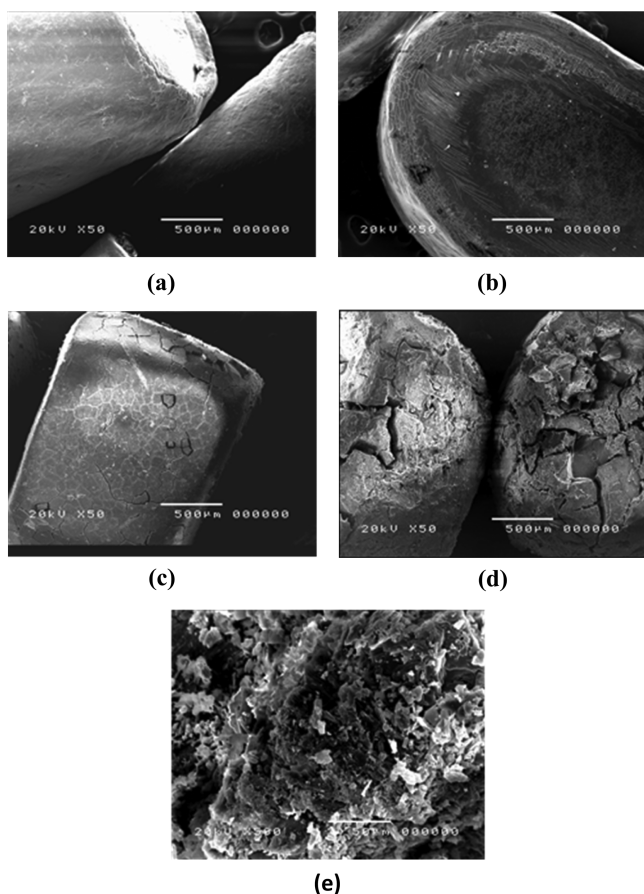


Figure 2. SEM images of (a) fresh PET; (b–d) residual PET pellets after reacting for (b) 15, (c) 30, and (d) 60 min; and (e) main product BHET.

from the cracks. Compared to the smooth surface of the PET pellets, BHET has a completely different, irregular morphology, as shown in Figure 2e.

The mechanism and proposed pathway of the glycolysis of PET catalyzed by the Lewis base ionic liquid [Bmim][OAc]^{47,48} are illustrated in Scheme 1; this process is a Lewis base catalytic reaction. When [Bmim][OAc] is added, it is

assumed that there exists a synergic effect between the cation and anion of this catalyst.⁴⁹ The cation of this ionic liquid, [Bmim]⁺, interacts with the carbonyl oxygen (C=O) in the ester of PET. The oxygen in the hydroxyl group of ethylene glycol then attacks the carbon cation of the ester group, forming a tetrahedral intermediate. Meanwhile, the anion of the catalyst, [OAc][−], interacts with the hydrogen in the hydroxyl group of ethylene glycol, resulting in the oxygen of ethylene glycol becoming more negative and better able to attack the carbon cation of the ester group, breaking the C–O bond in PET. Last, the hydrogen in ethylene glycol leaves and combines with [OAc][−] to form acetic acid as an intermediate. The acetic acid then partially dissociates to give H⁺ and an acetate ion. In this case, the EG molecule and carbon are connected, a new ester group is formed, and the chain of PET is cleaved. The electrons on the oxygen in —O(Bmim⁺) then transfer, forming C=O. The acyl oxygen cleaves, and the —OCH₂CH₂— group leaves, combining with H⁺ to form HOCH₂CH₂—.

The depolymerization of PET proceeds step-by-step, and oligomers, dimers, and monomers are generated sequentially. With the progress of the reaction, an increasing number of BHET monomers emerge, which can polymerize into dimers or oligomers, creating a chemical equilibrium.²⁷

Kinetics of the Glycolysis of PET. In studies of the depolymerization kinetics of polymers, the reaction order is usually considered to be first-order.^{50–54} Therefore, the glycolysis of PET in the presence of [Bmim][OAc] was initially assumed to be controlled by the first-order kinetic equation

$$\frac{d(C_{\text{PET}})}{dt} = -kC_{\text{PET}} \quad (4)$$

where k is the rate constant of the reaction and C_{PET} is the concentration of poly(ethylene terephthalate) at time t

$$C_{\text{PET}} = C_{\text{PET}0}(1 - X) \quad (5)$$

where X is the conversion of PET. Equation 4 can then be written as follows

$$\frac{dX}{dt} = k(1 - X) \quad (6)$$

Equation 6 was integrated over time to give

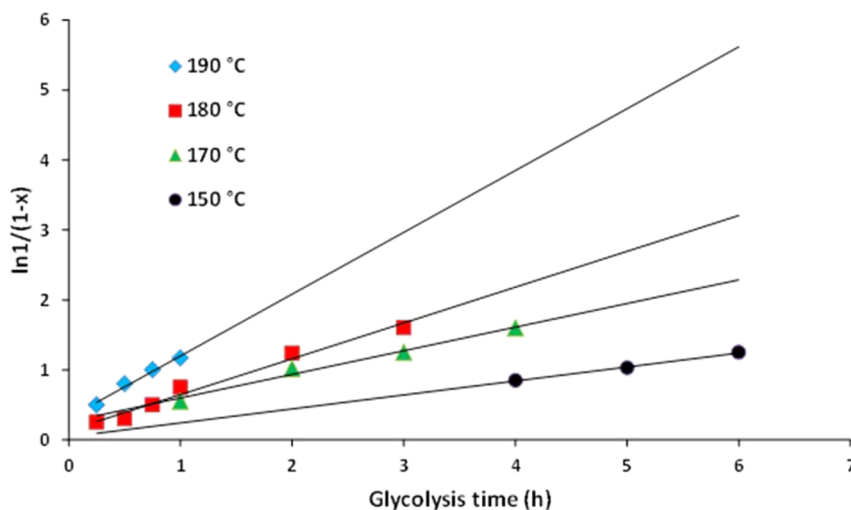


Figure 3. Effect of reaction temperature on the glycolysis rate of PET in [Bmim][OAc].

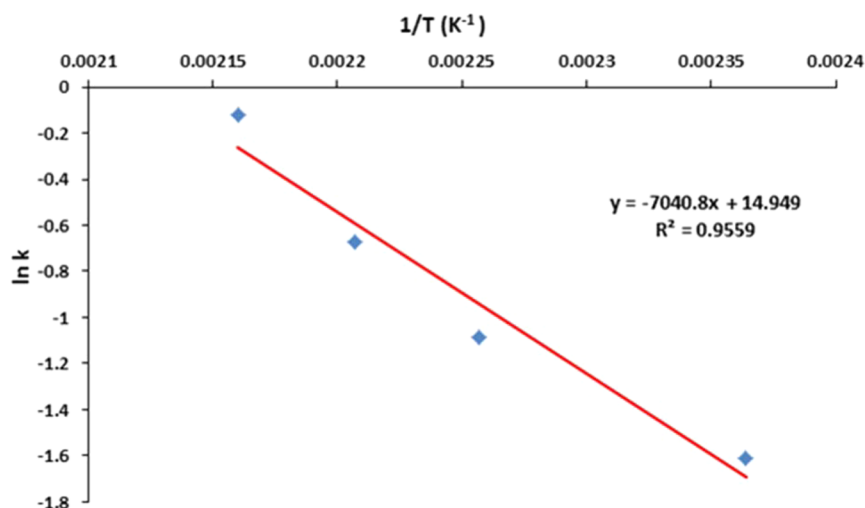


Figure 4. Arrhenius plot of the rate constant of the glycolysis of PET in [Bmim][OAc].

$$\ln \frac{1}{1-X} = kt \quad (7)$$

The effect of the reaction temperature on the glycolysis rate of PET in the presence of [Bmim][OAc] is shown in Figure 3, and the linear regression results are reported in Table S8 (Supporting Information). All of the linear correlation coefficients in Table S8 (Supporting Information) are higher than 0.97, which indicates that $\ln[1/(1-X)]$ was proportional to the reaction time at different temperatures and that this process followed a first-order kinetic reaction. The rate of the glycolysis of PET in the ionic liquid was found to be proportional to the PET concentration. The straight lines in Figure 3 have slopes of 0.2, 0.338, 0.5123, and 0.884 h^{-1} , which correspond to the rate constants of the glycolysis reaction at 150, 170, 180, and 190°C , respectively.

Using these values of the rate constant, the activation energy (E_a) was obtained from the relation

$$\ln k = \ln A - \frac{E_a}{RT} \quad (8)$$

where A is the pre-exponential factor, R is the gas constant (8.31 J/k mol), and T is the temperature in Kelvin. The activation energy for this reaction, calculated from the slope of the Arrhenius plot shown in Figure 4, was 58.53 kJ/mol .

CONCLUSIONS

In summary, it has been demonstrated that [Bmim][OAc] can behave as an efficient and ecofriendly catalyst for the glycolysis of PET in ethylene glycol. The results show that the reaction temperature is a critical factor in this process. Using [Bmim][OAc] (1.0 g) with 20 g of EG and 3.0 g of PET, it was found that only 3 h of glycolysis at 190°C was sufficient for the conversion of PET to reach 100% and the yield of BHET to reach 58.2%. The changes in pH in the glycolysis of PET were interpreted in accordance with the proposed mechanism. The synergistic effect of the cation and anion in the Lewis base ionic liquid [Bmim][OAc] facilitates the attack of the oxygen in ethylene on the carbon cation of the ester group. The acetate-based ionic liquid [Bmim][OAc] might have the potential to substitute traditional compounds used to catalyze the glycolysis of PET in industrial production, as it can be reused up to six times. In addition, the kinetics of the

reaction was also investigated, with the results indicating that the glycolysis of PET is a first-order reaction with an activation energy of 58.53 kJ/mol .

ASSOCIATED CONTENT

Supporting Information

EI-MS, FTIR, and ^1H NMR spectra of [Bmim][OAc]. Catalytic effects of different ionic liquids on the glycolysis of PET, effects of reaction conditions on the glycolysis reaction of PET using [Bmim][OAc], recycling of residual EG and [Bmim][OAc], and linear regression results. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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